Lakshya JEE 2.0 2025

Maths

Matrices

DPP: 3

Q1 If $A=[a_{ij}]$ is a skew-symmetric matrix of order n, then a_{ii} is equal to:
(A) 0 for some i
(B) 0 for all $\mathrm{i}=1,2,\ldots,\mathrm{n}$
(C) 1 for some i
(D) 1 for all $\mathrm{i}=1,2,\ldots\ldots,\mathrm{n}$

Q2 If A is skew-symmetric of order 3×3, then trace of A is _____.
(A) 0 (B) 1
(C) 2 (D) 3

Q3 The maximum number of different elements required to form a symmetric matrix of order 6 is
(A) 15 (B) 17
(C) 19 (D) 21

Q4 If A and B are symmetric matrices of the same order, then
(A) AB is a symmetric matrix
(B) $\mathrm{A}-\mathrm{B}$ is a skew-symmetric matrix
(C) $\mathrm{AB}+\mathrm{BA}$ is a symmetric matrix
(D) $\mathrm{AB}-\mathrm{BA}$ is a symmetric matrix

Q5 If A is a symmetric matrix and B is a skew-symmetric matrix such that $A+B=\begin{bmatrix}2 & 3\\ 5 & -1\end{bmatrix}$, then AB is equal to
(A) $\begin{bmatrix}-4 & -2\\ -1 & 4\end{bmatrix}$
(B) $\begin{bmatrix}4 & -2\\ -1 & -4\end{bmatrix}$
(C) $\begin{bmatrix}4 & -2\\ 1 & -4\end{bmatrix}$
(D) $\begin{bmatrix}-4 & 2\\ 1 & 4\end{bmatrix}$

Q6 If $\mathrm{A}_1,\ \mathrm{A}_2,\ \mathrm{A}_3,\ldots,\mathrm{A}_{2n-1}$ are n skew-symmetric matrices of same order, then $\mathrm{B}=\sum_{r=1}^{n}(2r-1)(A_{2r-1})^{2r-1}$ will be
(A) symmetric
(B) skew-symmetric
(C) neither symmetric nor skew-symmetric
(D) data not adequate

Q7 If $A=\begin{bmatrix}6 & 8 & 5\\ 4 & 2 & 3\\ 9 & 7 & 1\end{bmatrix}$ is the sum of a symmetric matrix B and skew -symmetric matrix C, then B is equal to
(A) $\begin{bmatrix}6 & 6 & 7\\ 6 & 2 & 5\\ 7 & 5 & 1\end{bmatrix}$
(B) $\begin{bmatrix}0 & 2 & -2\\ -2 & 5 & -2\\ 2 & 2 & 0\end{bmatrix}$
(C) $\begin{bmatrix}6 & 6 & 7\\ -6 & 2 & -5\\ -7 & 5 & 1\end{bmatrix}$
(D) $\begin{bmatrix}0 & 6 & -2\\ 2 & 0 & -2\\ -2 & -2 & 0\end{bmatrix}$

Q8 Let $A=\begin{pmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{pmatrix},(\alpha\in R)$ such that $A^{32}=\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}$ then a value of α is:
(A) 0 (B) $\pi/16$

(C) $\pi/32$ (D) $\pi/64$

Q9 Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is:

(A) 6 (B) 4

(C) 1 (D) 12

Q10 If $A=\begin{bmatrix}2 & 0 & 1\\ 2 & 1 & 3\\ 1 & -1 & 0\end{bmatrix}$ and $f(x)=x^2-5x+6$, then the value of $f(A)$ is:

(A) $\begin{bmatrix}0 & -1 & 3\\ 1 & 1 & -10\\ 5 & 4 & 4\end{bmatrix}$

(B) $\begin{bmatrix}0 & 0 & -3\\ 1 & 2 & 10\\ 6 & 4 & 1\end{bmatrix}$

(C) $\begin{bmatrix}1 & -1 & -3\\ -1 & -1 & -10\\ -5 & 4 & 4\end{bmatrix}$

(D) $\begin{bmatrix}1 & 2 & 3\\ -1 & -1 & 0\\ 4 & -5 & 4\end{bmatrix}$

Q11 Matrix A has m rows and $\mathrm{n}+5$ columns, matrix B has m rows and $11-n$ columns. If both AB and BA exist, then:

(A) AB and BA are square matrices

(B) AB and BA are of order 8×8 and 3×13, respectively

(C) $\mathrm{AB}=\mathrm{BA}$

(D) None of these

Q12 The total number of matrices $A=\begin{pmatrix}0 & 2y & 1\\ 2x & y & -1\\ 2x & -y & 1\end{pmatrix}(x,y\in R, x\neq y)$ for Which $\mathrm{A}^{\mathrm{T}}\mathrm{A}=3\mathrm{I}_3$ is:

(A) 3 (B) 4

(C) 2 (D) 6

Q13 If $\mathrm{A}=\begin{bmatrix}i & -i\\ -i & i\end{bmatrix}$ and $\mathrm{B}=\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$, then A^8 equals

(A) $128\ \mathrm{B}$

(B) $132\ \mathrm{B}$

(C) $116\ \mathrm{B}$

(D) $8\ \mathrm{B}$

Q14 If $A=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$, then $A^T+A=I_2$, if

(A) $\theta=n\pi, n\in Z$

(B) $\theta=(2n+1)\pi/2, n\in Z$

(C) $\theta=2n\pi+\pi/3, n\in Z$

(D) None of these

Q15 If $\mathrm{A}=[a_{ij}]$ is a square matrix of even order that $a_{ij}=i^2-j^2$, then

(A) A is a skew-symmetric matrix and $|\mathrm{A}|=0$

(B) A is symmetric matrix and $|\mathrm{A}|$ is a square

(C) A is symmetric matrix and $|\mathrm{A}|=0$

(D) none of these

Q16 If A and B are symmetric matrices of the same order and $X=AB+BA$ and $Y=AB-BA$, then $(XY)^T$ is equal to

(A) XY (B) YX

(C) $-YX$ (D) None of these

Q17 Let $A=\begin{bmatrix}\tan\frac{\pi}{3} & \sec\frac{2\pi}{3}\\ \cot\left(2019\frac{\pi}{2}\right) & \cos(2020\pi)\end{bmatrix}$ and P is a 2×2 matrix such that $PP^T=I$, where I is an identity matrix of order 2. If $Q=PAP^{\mathrm{T}}$ and $R=[r_{ij}]_{2\times 2}=P^TQ^8P$, then

(A) $r_{11}=81$

(B) $r_{11}=27\sqrt{3}$

(C) $r_{11}=4\sqrt{3}$

(D) $r_{11} = -\sqrt{3}$

Q18 Let A and B are the two non-singular square matrices, A^T and B^T are the two transpose matrices of A and B respectively, then which of the following are correct?

(A) B^TAB is symmetric matrix if A is symmetric.

(B) B^TAB is symmetric matrix if B is symmetric

(C) B^TAB is skew symmetric matrix for every matrix A.

(D) B^TAB is skew symmetric matrix if A is skew symmetric

Answer Key

Q1	(B)	Q10	(C)
Q2	(A)	Q11	(A)
Q3	(D)	Q12	(B)
Q4	(C)	Q13	(A)
Q5	(B)	Q14	(C)
Q6	(B)	Q15	(D)
Q7	(A)	Q16	(C)
Q8	(D)	Q17	(A)
Q9	(B)	Q18	(A, D)

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